

Building The Foundation For HARMLESS PEST CONTROL Using Electronics

Electronic pest control is the name given to electrically-powered devices designed to repel or eliminate pests. This article takes a look at such devices



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With lifestyles moving towards comfortability-to-harmful levels and pseudo-cleanliness, agriculture and living places of humans have come under drastic attack from pesticides and weedicides to control various types of pests and weeds. Most pesticides and weedicides contain harmful chemicals, which can have a detrimental effect on the well-being of the society and nature.

People in urban as well as rural areas are making excessive use of various pesticides and weedicides in their daily lives in general, and in agricultural and food items in particular. Even small and marginal farmers, fruit and vegetable growers, and most modern house owners want to be free from weeds and pests like mosquitoes, cockroaches and rodents.

All types of agricultural products and food items enter the human body in one form or the other. Even pesticides and weedicides that are used to protect crop residues, grasses and plants enter the human body through agricultural and animal products.

Use of pesticides has resulted in devastating effects on humans, animals and biodiversity. Various species of birds, aquatic organisms and animals have already become extinct due to excessive use of pesticides. Humans are also suffering adverse effects, which are weakening their immune system and, hence, leading to various diseases.

Electronic pest control

Electronics has influenced almost all aspects of life from medical, entertainment and safety to sports, education, defence and so on. To control or avoid the adverse effects of harmful pesticides on humans, animals and biodiversity, electronic pest control is a desired technology.

An electronic pest control system does not use any chemicals, messy traps, poisons or other potentially harmful substances. In fact,

electronic pest control does not kill pests, mosquitoes and rodents (as killing is not desired for sustainable biodiversity) but only repels them. Hence, an electronic pest repellent is an environment- and human-friendly pest control technology.

Moreover, latest electronic pest and mosquito machines are well-equipped with new air-purifying technology. These are usually based on the use of either electromagnetic, ionic or ultrasonic technology, or all in one. Electronic pest control devices based on these principles have many user-friendly features and capacities.

However, there is a need to make these devices versatile in terms of capacity (bringing these out of homes to fields) and cost-effectiveness. This article discusses the fundamentals of different electronic pest control technologies based on electromagnetic, ionic and ultrasonic principles, and the latest development towards the same.

Electronic pest control techniques

Electronic pest control is the name given to any of the several types of electrically-powered devices designed to repel or eliminate pests. This technique usually makes use of electromagnetic waves, ultrasonic waves and ions, either by themselves or clubbed together.

Ultrasonic repellents. Ultrasonic devices operate by emitting short-wavelength, high-frequency sound waves that are too high in pitch to be heard by the human ear (generally, frequencies greater than 20,000Hz are termed as ultrasonic waves). Animals like bats, dogs and rodents can hear well into the ultrasonic range.

There are insects like locusts/grasshoppers and moths/lacewings that can detect frequencies of 50kHz -100kHz and as high as 250kHz, respectively. Such high-frequency waves are also produced by insect-hunting bats. Some